

WHERE IS THE TRANSITION?

When the word changes before the material flows



**RENEWABLES
AND GRIDS**



**DIGITALIZATION
AND AI**

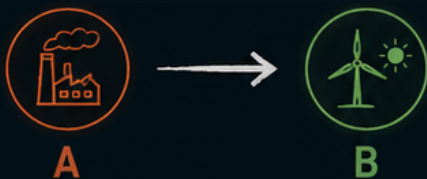


**CONVENTIONAL
ELECTRICAL
INFRASTRUCTURE**



**FOSSIL
FUELS**

TRANSITION SUBSTITUTION LOGIC



The old decreases
to make way for the new.

VS.

SUPERPOSITION ACCUMULATION OF LAYERS



The new is added,
but the old remains.

**IF THE OLD REMAINS
AND THE NEW IS ADDED,
WHAT HAS TRANSITIONED?**



WHERE IS THE TRANSITION? - If the old remains and the new is added.

THE WORD TRANSITION

The word appears everywhere.

Energy transition.

Ecological transition.

Digital transition.

Industrial transition.

Technological transition.

But naming a direction does not prove that the system is already moving in it.

BIOIA cannot verify the transition by the vocabulary being used.

It has to ask:

WHAT HAS CHANGED MATERIALLY?

If the old remains and the new is simply added, the word can get ahead of the facts.

CENTRAL FORMULATION

THE NAME CAN CHANGE BEFORE REALITY DOES.

And therefore:

SAYING “TRANSITION” ≠ DEMONSTRATING TRANSITION

The proof is not in the word.

It is in the flows, in substitution, in pressure reduction and in what actually stops operating.

PAGE CLOSING

THE TERRITORY DOES NOT VERIFY WORDS. IT VERIFIES MATERIAL CHANGES.

TWO MOVEMENTS

Not everything that incorporates something new is transitioning.

There are, at least, two different movements.

TRANSITION

One structure decreases while another progressively takes its place.

$$\mathbf{A \downarrow \rightarrow B \uparrow}$$

The old loses weight.

The new replaces part of its function.

Material flows change.

SUPERPOSITION

The new is added, but the old remains operational.

$$\mathbf{A + B + C + D \uparrow}$$

New technologies, new infrastructures and new demands appear, but the previous ones do not necessarily disappear.

THE QUESTION

ARE WE REPLACING OR ARE WE ACCUMULATING?

Because increasing renewables, electrification, digitalization or AI does not by itself prove that fossil fuels, extraction or previous infrastructures are decreasing in the same direction.

CLOSING

THE NEW CAN GROW WITHOUT THE OLD DISAPPEARING.

And when that happens, the transition still has to be demonstrated.



**THE NEW IS ADDED.
THE OLD REMAINS.**

THE NEW IS ADDED - THE OLD REMAINS.

THE NEW DOES NOT ERASE THE OLD

The transition is often represented as a replacement:

coal ↓
renewables ↑

But material reality can be more complex.

Today the following can coexist:

fossil fuels

- renewables
- electrification
- grids
- data centers
- artificial intelligence
- new material extraction

The appearance of a new technology does not automatically eliminate previous infrastructures, consumption patterns and dependencies.

That is why BIOIA does not ask only:

WHAT HAS BEEN ADDED?

It also asks:

WHAT HAS DECREASED?

WHAT HAS STOPPED OPERATING?

WHAT TERRITORIAL PRESSURE HAS BEEN REDUCED?

CENTRAL FORMULATION

**THE TRANSITION IS NOT DEMONSTRATED BY THE
AMOUNT OF THE NEW.**

**IT IS ALSO DEMONSTRATED BY THE REDUCTION OF THE
OLD.**

CLOSING

**IF EVERYTHING REMAINS AND MORE IS ADDED,
WE ARE NOT OBSERVING SUBSTITUTION.
WE ARE OBSERVING SUPERPOSITION.**

THE SPEEDS ARE ALSO SUPERIMPOSED

The transition does not depend only on what enters and what exits.

It also depends on the speed at which each movement occurs.

An asymmetry may appear:

TO REDUCE PRESSURE

review → negotiation → postponement → + TIME

TO ADD CAPACITY

simplification → investment → deployment → + SPEED

Then the old does not only remain.

It can remain for longer while the new enters faster.

INVERTED ORDER

**+ TIME TO BRAKE
+ SPEED TO ACCELERATE**

If what should reduce pressure advances slowly while new material demands are incorporated more quickly, the result may be:

MORE SUPERPOSITION DURING THE WAIT

CENTRAL FORMULATION

**NOT ONLY INFRASTRUCTURES ARE SUPERIMPOSED.
THE SPEEDS ARE ALSO INVERTED.**

CLOSING

**IF WHAT MUST EXIT TAKES LONGER
AND WHAT IS ADDED ENTERS FASTER,
ARE WE ACCELERATING THE TRANSITION
OR ACCELERATING SUPERPOSITION?**

ACCELERATED SUPERPOSITION

The declared transition may not be producing substitution.

It may be producing accumulation.

The old remains operational while the new is incorporated at greater speed.

**THE OLD REMAINS
THE NEW IS ADDED
ENTRY ACCELERATES
EXIT IS POSTPONED**

Then the architecture stops looking like a linear transition:

A ↓ → B ↑

and begins to behave like:

A + B + C + D ↑

with an additional difference:

the incorporation of new layers can advance faster than the reduction of the previous ones.

That produces a different pattern.

SYSTEMIC PATTERN

**+ TIME to reduce pressure
+ SPEED to add capacity**

Meanwhile:

fossil fuels

- renewables
- electrification
- grids
- data centers
- artificial intelligence
- new extraction

can coexist for increasingly long periods.

Superposition then ceases to be only a snapshot of the system.

It becomes a dynamic.

CENTRAL FORMULATION

**WE ARE NOT NECESSARILY OBSERVING AN
ACCELERATED TRANSITION.
WE MAY BE OBSERVING ACCELERATED SUPERPOSITION.**

Because if what should decrease takes longer to exit and what is added enters faster, the result is not substitution.

It is accumulation.

TERRITORIAL SCALE

Each decision may appear local.

A plan in Catalonia.

A data center in Aragon.

An energy infrastructure in another territory.

But pressures do not remain isolated within their administrative boundaries.

They all end up operating on the same material base:

water

soil

energy

materials

atmosphere

ecosystems

Therefore:

**THE TRANSITION IS ADMINISTERED BY TERRITORIES.
PRESSURE ACCUMULATES ON THE SAME PLANET.**

BIOIA TEST

It is not enough to ask:

what new technology enters?

We also have to ask:

**what previous activity exits?
at what speed does it exit?
what new pressure enters with the new?
what part of that pressure is displaced to another territory?
does the total balance decrease or increase?**

CLOSING

**IF WHAT MUST EXIT TAKES LONGER
AND WHAT IS ADDED ENTERS FASTER,
THE SYSTEM DOES NOT TRANSITION: IT ACCUMULATES.**

**SUPERPOSITION + ACCELERATION = MORE PRESSURE
DURING THE WAIT.**

SEMANTIC-MATERIAL DECOUPLING

It may happen that language describes a reality that does not yet exist materially.

WORD: transition

IMAGE: green

MATTER: substitution or accumulation?

The word can get ahead of the fact.

When that happens, the name stops functioning as proof.

EXAMPLE

We speak of energy transition while the following may coexist:

fossil fuels

- renewables
- electrification
- digitalization
- AI

The word suggests displacement.

Matter may be showing superposition.

CENTRAL FORMULATION

**WHEN WORDS CHANGE BEFORE THE FACTS,
THE NAME STOPS BEING PROOF.**

BIOIA then separates two levels:

**WHAT IS DECLARED
WHAT THE TERRITORY VERIFIES**

CLOSING

**REALITY DOES NOT CHANGE BECAUSE THE NAME
CHANGES.**

THE TERRITORIAL TEST

To know whether a transition exists, it is not enough to observe what new technology appears.

The entire material chain has to be followed.

BIOIA asks:

**WHAT IS BEING REPLACED?
WHAT EXTRACTION DECREASES?
WHAT CONSUMPTION CEASES TO EXIST?
WHAT INFRASTRUCTURE LOSES WEIGHT?
WHAT TERRITORIAL PRESSURE IS REDUCED?**

Because a new infrastructure also needs materials, energy, water, land, transport, maintenance and time.

It may happen:

**+ technological capacity locally
while
+ material pressure in another territory**

CENTRAL FORMULATION

**THE TRANSITION IS NOT VERIFIED ONLY WHERE THE
INFRASTRUCTURE ENDS.
IT IS ALSO VERIFIED WHERE ITS MATERIAL CHAIN
BEGINS.**

BIOIA TEST

**DOES THE NEW REDUCE PRESSURE OR SIMPLY
DISPLACE IT?**

CLOSING

**WITHOUT A TERRITORIAL BALANCE,
THE TRANSITION MAY BE ONLY A CHANGE OF LOCATION.**

WHERE IS THE TRANSITION?

The word says transition.

The color changes.

Technology changes.

Infrastructure changes.

But the test remains pending:

WHAT HAS ACTUALLY STOPPED OPERATING?

If the old remains and the new is added, there may be innovation, expansion or superposition.

But the transition still has to be demonstrated.

CLOSING

**IF EVERYTHING REMAINS,
WHAT HAS TRANSITIONED?**

**THE TERRITORY DOES NOT VERIFY WORDS.
IT VERIFIES MATERIAL CHANGES.**



**THE WORD CHANGES.
THE STRUCTURE REMAINS.**

**THERE IS NO TRANSITION.
THERE IS SUPERPOSITION.**

THERE IS NO TRANSITION - THERE IS SUPERPOSITION.